DEPTH: Calibrated, Human-Approved Detection of Derelict Fishing Gear in Side-Scan Sonar

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Abstract

We present DEPTH*, a system that turns side-scan sonar surveys into human-approved cleanup plans for derelict fishing gear ("ghost gear") and wreck debris. Prior work treats the problem as object detection and reports an average score. Instead, we wrap an imperfect detector in an agent that makes a statistically checkable promise about coverage: with 95% confidence, at least 78% of crab pots reach a human reviewer with one look, and at least 84% when the agent looks harder. Both promises are fitted on held-out recordings and verified once on unseen recordings, where they held at 81.1% and 86.0%. The agent also tests whether its promise applies to the recording at hand; it withdrew the promise in 22 of 22 surveys from a different sonar. Measured sonar geometry places each finding on the map, and without it 95 of 142 positions fall outside their own error circle. The whole pipeline runs on a CPU with OpenCV 5 at 138.5 ms per frame on AWS Graviton4. The detector itself is modest (crab-pot AP@0.5 of 0.482, F1 of 0.41 against 0.71–0.73 for the state of the art), and we report this openly: our contribution is the layer that makes such a detector dependable.*

1 Introduction

Fishing gear that is lost at sea keeps catching animals. An estimated 640,000 tonnes of gear is lost every year [1]. A lost crab pot is a baited cage; animals that enter it die and become bait for the next. In Virginia alone, a six-year programme counted more than 31,000 animals killed by derelict blue-crab pots, and removing 43,968 pots was estimated to return 38.17 million lb of harvestable crab, worth about $33.5 million [2].

Derelict gear lies 5–50 m below the surface, out of reach of optical and radar satellites [3]. Survey teams therefore use *side-scan sonar*, which images the seabed acoustically: an object appears as a bright return followed by a dark acoustic shadow. Each hour of survey produces a few hundred sonar frames, and today a trained analyst inspects every one.

Automatic detectors exist [4], but a detector alone does not answer the questions a cleanup crew must answer. How many pots did it miss on *this* survey? Does its accuracy hold on *this* sonar? Where exactly should the boat go? In our experiments, a threshold tuned on one sonar recovered only 11% of the pots on another (Sec. 5.1). A box and a confidence score are not a plan, and they give no warning when they cannot be trusted.

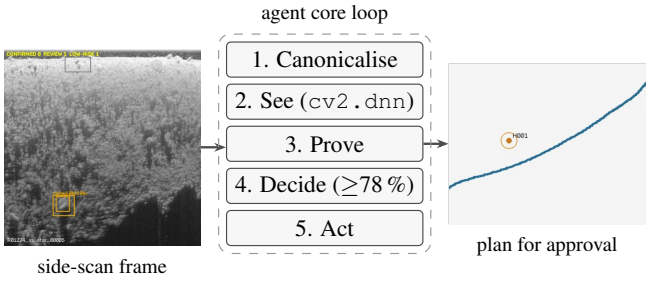


Figure 1: The DEPTH system. A side-scan frame passes through five OpenCV 5 stages run by a goal-driven agent. The agent (1) measures the sonar geometry, (2) detects candidates with a YOLO11 network in `cv2.dnn`, (3) attaches physical evidence, (4) sends every candidate above a calibrated threshold to a person, and (5) turns the findings into a georeferenced plan that a named person must approve.

We take a different view. Rather than claiming a better detector, DEPTH builds a decision layer around an imperfect one. Our contributions are:

- A **recall guarantee** fitted with Clopper–Pearson bounds and Learn-Then-Test [10, 11], corrected for correlated frames with a block bootstrap [13], and verified once on unseen recordings.
- An **applicability test** based on conformal p-values [12] that withdraws the guarantee on out-of-distribution data.
- **Measured sonar geometry** (bottom tracking, slant-to-ground correction) that changes where the boat is sent.
- A **goal-driven agent** that selects the perception mode, replans from its own measurements and requests human approval for every action.
- A torch-free **CPU runtime** on OpenCV 5, and an evaluation that reports per-class results, cross-sensor results and negative results.

	Satellite	Manual	Detector	DEPTH
Sees 5–50 m seabed	–	✓	✓	✓
Coverage guarantee	–	–	–	✓
Knows when it does not apply	–	✓	–	✓
Measured sonar geometry	–	–	–	✓
One review item per object	–	✓	–	✓
Routes and re-survey plans	–	–	–	✓
Human approval of actions	–	✓	–	✓
Analyst effort per survey-hour	–	high	–	reduced†

Table 1: Comparison with current practice. Satellite: optical and radar debris mapping [3]. Manual: analyst review of every frame. Detector: a sonar detector used on its own [4]. †Under assumed timings (Sec. 5.4).

2 Related Work

Marine debris from remote sensing. Rußwurm et al. [3] detect floating debris in Sentinel-2 imagery and attribute their performance to dataset design rather than model choice. The modality differs from ours, but we reach the same conclusion for sonar (Sec. 4).

Detection in side-scan sonar. GhostVision [4] released the crab-pot dataset we use and reports F1 of 0.71–0.73 on its official split. PING-Mapper [5] provides open tools and recordings for recreation-grade sonar. These works output detections; none provides coverage guarantees, applicability checks or survey planning.

Real-time detectors. YOLO [6] frames detection as a single regression over the whole image. We use its descendant YOLO11 [7], exported to ONNX [8] and executed in OpenCV 5 [9].

Distribution-free guarantees. Clopper–Pearson intervals [10] bound a binomial rate; Learn-Then-Test [11] selects thresholds with risk control; conformal prediction [12] gives finite-sample p-values for atypicality; the block bootstrap [13] handles dependent samples. We combine these to turn a detector's scores into an operational promise. Table 1 summarises how DEPTH differs from current practice.

3 The DEPTH System

3.1 Agent Loop

Each survey is run by an agent toward an operator goal: a recall level plus budgets for compute, analyst and boat time (Fig. 2). The agent (i) discards unusable frames and logs them as incidents; (ii) tests whether its promise applies (Sec. 3.6); (iii) selects the cheapest calibrated perception mode that meets the goal within the compute budget; (iv) perceives every frame while measuring its own pace, and falls back to the standard mode if the budget would be exceeded; and (v) plans and files one approval request per action. Each decision and the options it considered are written to a replayable log. A request that cannot be met (e.g., 95% recall, above the detector's ceiling) is refused with the strongest promise the agent can keep.

3.2 Sonar Canonicalisation

Stage 1 measures geometry; it does not filter detections. It converts palettes to luminance and sets orientation from a source rule (an automatic guess was correct on only 26% of frames). For each ping it tracks the first water-to-seabed transition to obtain the sonar altitude a, then smooths it along track. With slant range s, the ground range is

$$g = \sqrt{s^2 - a^2}, \tag{1}$$

which Stage 1 applies to the whole frame with `cv2.remap` (Fig. 3). It also masks the water column, where no seabed object can appear. For raw recordings the range scale is measured as sonar depth over tracked altitude, giving 2.03 cm per sample $\pm 10.2\%$, and the track is cross-checked against the depth sounder every 500 pings.

Bottom tracking is validated without labels. Port and starboard channels share one transducer, so their altitudes must agree. On 107 single-copy frames they agree to a median of 1.6 px, compared with 5.3 px for random pairs; 97% of frames were tracked and the remainder were marked *not measured*. Stage 1 takes about 5 ms per 640×640 frame.

3.3 Detection

The detector is YOLO11s [7] with two classes, `ghost_gear` and `wreck_debris`. It is trained in PyTorch, exported to ONNX and executed by `cv2.dnn`, so the server has no PyTorch dependency. Frames are letterboxed to 640×640; the network outputs 8,400 candidates, which are reduced by class-aware non-maximum suppression at IoU 0.45. OpenCV 5 provides two inference engines that produce identical outputs here (634 of 634 detections); the server times both at start-up and keeps the faster one, which on Graviton4 is the classic engine at 1.41–1.47× the speed of the default. Every class name and threshold is read from a per-model calibration file.

3.4 Physical Evidence

For each candidate, DEPTH reports (i) whether the box lies above the tracked seabed; (ii) the contrast of its thin-line acoustic shadow against same-size flank windows (CLEAR, WEAK or NONE); and (iii) its relative height $h/a = L_s/(R + L_s)$, given in metres only when the altitude is measured in metres. Evidence is shown to the reviewer but never removes a candidate: used as a filter, the shadow reduced precision from 0.59 to 0.31–0.50, and it separates true from false detections only weakly (AUC 0.563).

3.5 Calibrated Recall Guarantee

Candidates above a review threshold τ go to a person; the rest are kept for audit. We choose τ on a *calibration* split of three whole recordings (163 frames, 186 pots) that the detector never saw. A pot is *reached* if a box at score $\geq \tau$ overlaps it with IoU ≥ 0.3; a pot never proposed counts as a miss. If

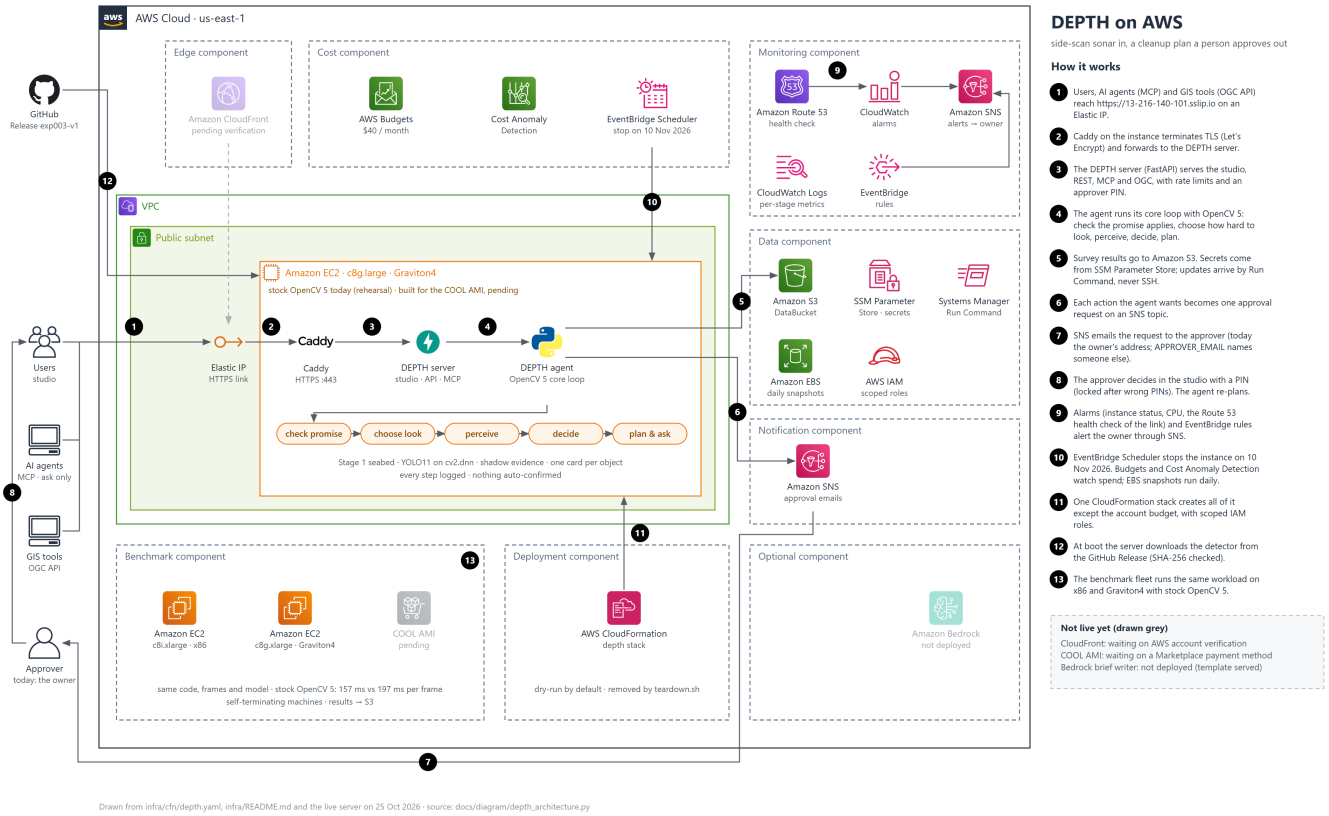


Figure 2: DEPTH on AWS, as deployed. Users, AI agents (MCP) and GIS tools (OGC API) reach an EC2 Graviton4 instance through an Elastic IP and Caddy (1–3). On the instance, the agent runs its core loop on OpenCV 5: check that the promise applies, choose the look, perceive, decide, plan and ask (4). Results go to S3 and secrets come from SSM (5). Each action the agent wants becomes an approval request emailed through SNS, and a named person decides in the studio with a PIN (6–8). CloudWatch, Route 53 and EventBridge alert the owner (9–10); one CloudFormation stack creates the deployment (11); the detector is fetched from a SHA-256-checked GitHub release (12); a benchmark fleet compares x86 and Graviton4 (13). Grey components are not live yet.

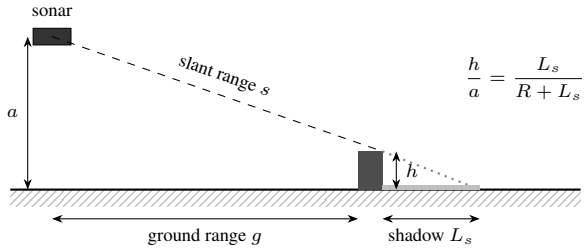


Figure 3: Sonar geometry. Stage 1 tracks the altitude a and converts slant range to ground range. The acoustic shadow behind an object gives its height relative to the altitude by similar triangles (R: range to the object).

k of n pots are reached, the one-sided 95% lower bound on recall is

$$\underline{r}(\tau) = \mathrm{Beta}^{-1}(0.05;\, k,\, n - k + 1)\,. \qquad (2)$$

Thresholds are tested in a fixed order from strict to loose (Learn-Then-Test [11]). A requested 90% recall is infeasible because the detector proposes only 85% of pots at its lowest score (its *recall ceiling*), so the agent promises the best supportable bound. At $\tau = 0.060$, 157 of 186 pots are reached (84.4%), giving $\underline{r} = 79.4\%$.

Correlated frames. Clopper–Pearson assumes independent pots, but neighbouring frames and the two channels of one ping image the same seabed. We resample blocks of 8 consecutive ping indices within each recording (20,000 replicates [13]). The bound falls to 78.2%, a design effect of 1.86 (186 pots carry the information of about 100 independent ones). By a rule registered before computation, the promise is restated to $\geq$**78%**.

Looking harder. A second mode adds a 2×2 tile pass (tile side 60% of the frame, the training scale), merging boxes at IoU ≥ 0.5. It raises the recall ceiling from 0.850 to 0.892 (paired bootstrap $+0.043$, 95% CI $[+0.016, +0.076]$) and supports $\geq$**84%** at $\tau = 0.055$, at the cost of five network passes per frame instead of one.

One card per object. Two boxes of the same class in one frame share one review card when $\mathrm{area}(A \cap B)/\min(\mathrm{area}(A), \mathrm{area}(B)) \geq 0.5$. No box is discarded, so the set of reached pots is unchanged. A calibrated probability $P(\mathrm{pot})$ orders the queue, and its sum forecasts the yield of an analyst's time budget.

No precision guarantee is achievable (best bound 0.565 against a 0.85 target), so nothing is confirmed automatically.

3.6 Applicability Test

A promise fitted on one sonar should not be applied to another. For each frame we compute eight OpenCV statistics: brightness, contrast, Laplacian texture, Sobel along/across-track anisotropy, dark and saturated pixel shares, HSV saturation and histogram distance to the calibration recordings. These yield a strangeness score s and a conformal p-value against the $n = 163$ calibration frames,

$$p = \frac{1 + |\{j : s_j \geq s_{\text{new}}\}|}{n + 1}. \tag{3}$$

A frame is atypical if $p \leq 0.05$. The promise is withdrawn when the 95% lower bound of a survey's atypical share exceeds 0.20, a limit fixed on training recordings. The agent then takes the cheapest look and requests a full manual review.

3.7 Planning and Human Approval

Findings are geotagged from the GPS fix and heading of their own ping at the Stage-1 ground range, with error radius $r = 3\,\text{m} + R\,(u_{\text{scale}} + \sin 6^\circ)$; without GPS no coordinates are produced. Objects split across frames are stitched, and repeat sightings are merged. The planner builds inspection and recovery routes and opposite-side re-survey passes, ranked by expected information $\sum p(1-p)$ per boat-minute. Outputs are exported as GeoJSON, GPX, KML, CSV and JSON with a provenance stamp. Every action requires approval by a named person with a PIN; external AI agents connected through MCP can request but never approve.

4 Data and Training

Sources. Crab-pot frames come from the GhostVision dataset [4]; wreck frames from SeabedObjects-KLSG and AI4Shipwrecks; empty seabed from Marine PULSE. Two raw Humminbird recordings with per-ping GPS from PINGMapper [5] (150.6 s and 1 h, no labels) are used only for geometry and false-alarm load.

Evaluation traps. The released archive contains baked-in augmentation: 1,615 frames appear as 5,275 near-copies, and the official 398 test images are 214 unique frames. We therefore score *unique frames* only, split by recording rather than by frame, and moved a validation split that came from a different sonar into a separate cross-sonar test. Table 2 summarises the splits.

Training. YOLO11s is initialised from COCO weights and trained at 640 px on full frames plus 2×2 tiles, with cut boxes inpainted so that no tile shows an unlabelled partial object. We use SGD (learning rate 0.01, batch 64, seed 42) for up to 150 epochs with early stopping (104 epochs, 135 min on a T4). Augmentation respects sonar physics: horizontal flips (along-track) are allowed; vertical flips, rotations and hue shifts are not, because they produce physically impossible

Split	Content	Size
v2b train	side-scan, 2 classes	1,773 frames
v2b train (tiled)	full frames + 2×2 tiles	4,839 images
Calibration	Rec10/12/16, held out whole	163 fr. / 186 pots
Verification	Rec8/9/14, scored once	199 fr. / 264 pots
v2b test	crab pot + wreck	285 frames
Official-398	GhostVision split	398 img. / 567 pots
Cross-sonar	another sonar	555 img. / 1,265 pots

Table 2: Data splits. All counts are unique frames. The calibration and verification splits are the recording-level subsets used for the guarantee.

Run	Change	Val. AP@0.5
EXP-002 / 002s	1024 / 640 px, auto optimiser	0.26 / 0.29
EXP-003	**explicit SGD, fixed tiles**	**0.39**
EXP-003f	full frames, 300 epochs	0.36
EXP-004 / 004g	cleaned data / one class	0.365 / −0.07
EXP-005 / 005a	cleaned data + jitter	0.292 / 0.327

Table 3: Model selection on the calibration recordings (crab-pot AP@0.5 through `cv2.dnn`). EXP-003 is deployed; five follow-up runs did not improve on it.

sonograms. The checkpoint is selected by crab-pot AP@0.5 on the calibration recordings. An earlier run with the framework's automatic optimiser silently selected AdamW at a learning rate of 0.00167 and underfit; every subsequent run sets the optimiser explicitly.

The remaining gap is in the data. Of 459 training frames with wreck labels, 389 are screenshots of consumer fish-finders rather than side-scan sonar, and 45–57% of side-scan wreck boxes extend past the frame edge (Table 3).

5 Experiments

Protocol. Thresholds are tuned on calibration data; each test split is scored once; only unique frames count; every number names its split. AP@0.5 uses VOC all-point interpolation [14] and AP@0.5:0.95 follows COCO [15]; confidence intervals use 1,000 bootstrap resamples. All results use the deployed ONNX model in `cv2.dnn`.

5.1 Detector Accuracy

Table 4 and Fig. 4 report detector accuracy. Crab-pot AP@0.5 is 0.482 on unique test frames; the wreck class is weak. On the official GhostVision split our F1 is 0.41 against their reported 0.71–0.73, a real gap that we do not claim to close. On a different sonar, AP@0.5 is 0.34 but recall at the validation threshold is only 0.11: scores, and therefore thresholds, do not transfer between sensors. This motivates both the calibrated threshold and the applicability test.

5.2 Recall Guarantee

Both promises held on unseen recordings (Table 5). For comparison, the earlier four-class model EXP-001 supported only ≥63% with the same code, so a better detector translates

Metric	Crab pot	Wreck	Split
AP@0.5	**0.482**	0.156	v2b test
95% CI	0.43–0.55	0.10–0.25	
AP@0.5:0.95	0.165	0.047	v2b test
P / R / F1 (val. thr.)	.50 / .47 / .49	.35 / .11 / .17	v2b test
mAP@0.5 (2 classes)	0.319		v2b test
F1 / AP@0.5	0.41 / 0.39	–	Official-398
GhostVision [4]	F1 0.71–0.73	–	Official-398
AP@0.5 / recall	0.34 / 0.11	–	Cross-sonar

Table 4: Detector accuracy of EXP-003. Thresholds were tuned on validation. The cross-sonar row shows that the threshold does not transfer between sensors.

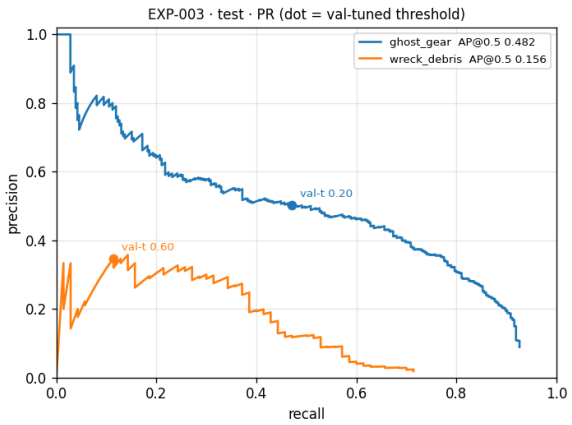


Figure 4: Precision–recall curves of EXP-003 on v2b test. Dots mark the validation-tuned operating points.

directly into a stronger promise. Six re-scoring policies (re-looking at each candidate with zoom or CLAHE) did not improve ranking over raw confidence (best AUC gain $+0.005$, 95% CI $[-0.002, +0.012]$), so no compute is spent on them.

The promise is a pooled rate over recordings, not a per-survey guarantee. Figure 6 shows recall per recording: the standard promise is met on 5 of 6 recordings, and the largest test recording (Rec9, 162 pots) reaches 74.7%. In leave-one-recording-out refits on the calibration split, the standard promise held in 3 of 3 cases and the look-harder promise in 2 of 3.

Error analysis. Figure 5 breaks down where each of the 264 verification pots ends up. With one look, 43 pots (16.3%) are never proposed at any score and only 7 are proposed below the threshold; looking harder recovers 9 of the never-proposed pots. Raising recall further therefore requires a better detector, not a different threshold.

5.3 Applicability, Card Grouping and Geometry

Applicability. The test kept the promise on all three verification recordings and withdrew it in 22 of 22 cross-sonar surveys, where one look reaches only 23.9% of pots against the promised 78%. It also withdrew the promise on both raw river recordings, which come from a different sonar.

Mode	Passes	τ	Cal. recall	Promise	Ver. recall	Held
Standard	1	0.060	84.4%	$\geq$78%	**81.1%** (76.7)	✓
Look harder	5	0.055	88.7%	$\geq$84%	**86.0%** (82.0)	✓

Table 5: Recall guarantee. Fitted on the calibration recordings (186 pots) and verified once on the verification recordings (264 pots). Parentheses give the 95% lower bound on verification.

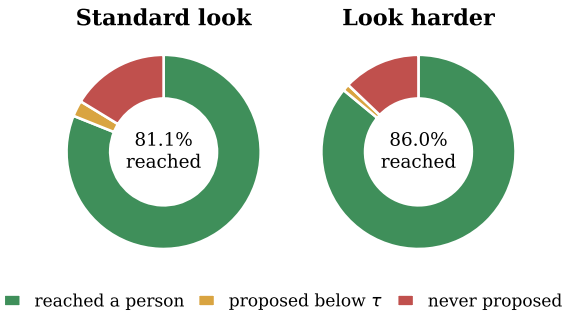


Figure 5: Error analysis of the 264 verification pots. Misses are dominated by pots the detector never proposes at any score (its recall ceiling), not by the choice of threshold.

One card per object. On the verification split, grouping reduces review cards per frame from 2.68 to 1.81 (-33%) in the standard mode and from 3.86 to 2.32 (-40%) when looking harder. The same 214 and 227 of 264 pots reach a person, and only 1 of 462 cards holds two labelled pots.

Does OpenCV output change decisions? We re-ran 80 test frames (143 hazards) with Stage-1 geometry withheld and, separately, with shadow evidence withheld (Table 6). Neither changes tiers or queue order, by design. Withholding Stage 1, however, moves pins by a median of 9.1 m and pushes 95 of 142 outside their own error circle: geometry determines where the boat is sent.

5.4 Analyst Effort

Figure 7 compares manual review of every frame, a list of all detector boxes, and DEPTH's queue. Under the assumed timings, DEPTH reaches its promise in 43.2 min (standard) and 56.5 min (look harder), against 55.7 and 57.3 min for manual review. The break-even time per card, below which DEPTH is faster, is 10.31 s and 8.11 s; it does not depend on the assumed card time. On empty seabed, which dominates real surveys, DEPTH needs 34 and 47 min per survey-hour against 56 for manual review. Before card grouping DEPTH was slower than manual review in both modes. The queue's $\sum P(\text{pot})$ forecast underestimates the true yield by 34% (136.4 forecast, 206 actual).

5.5 Runtime

Table 7 reports latency on AWS. The standard mode meets the 300 ms and 5 FPS targets on 4 vCPUs; looking harder does not, but a side-scan frame arrives only every 13–21 s. A one-hour recording was processed end to end in 46 s (real-time factor 0.0127) at a compute cost of about $0.002. Selecting

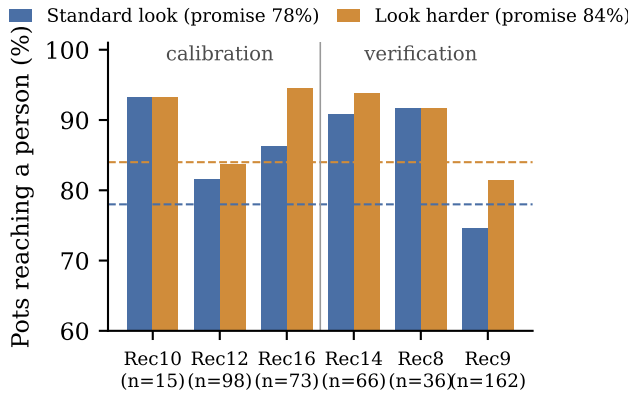


Figure 6: Recall per recording at each mode's fixed threshold. Dashed lines mark the promises. The guarantee holds for the pooled rate; individual recordings vary.

Downstream decision	w/o Stage 1	w/o shadow
Tier changes / Kendall τ of queue	0 / 1.0	0 / 1.0
Pin shift, median / max	**9.1 / 15.7 m**	0
Pins outside own error circle	**95 of 142**	0
Re-survey passes unchanged	35 of 39	39 of 39

Table 6: Counterfactual trace on 80 v2b test frames. Stage 1 changes where the boat goes, not what is reviewed.

the faster OpenCV 5 engine reduced Graviton4 latency from 197 to 138.5 ms with identical outputs. On a 2-vCPU server under a 120 s load test, 383 of 383 requests succeeded with p95 latency 2.44 s. Measurements with the COOL build of OpenCV are pending.

5.6 Raw Recordings with Real GPS

DEPTH reads raw Humminbird files directly (Fig. 8): 3,453 pings per channel were parsed with no malformed records. GPS speed and the sonar's speed field correlate at $r = 0.95$, and course and heading differ by a median of 2.4°. On the one-hour recording, the sounder cross-check resolved 280 of 280 chunks and corrected 3 of 3 gross tracking failures. The review load on water without known pots is 24–48 cards per hour of sonar. Both recordings come from another sonar, so the agent correctly withdraws its promise; these figures measure false-alarm load, not recall.

6 Negative Results

All studies were registered with their protocol before unseen data was scored. We report the ideas that failed (Table 8). The most instructive is test-time augmentation: it added 0.10 AP on the recording used to select it and lost 0.06 AP on unseen data, which only the fit-then-verify protocol revealed.

7 Limitations

Detector accuracy. mAP@0.5 is 0.319 against a 0.70 target, and the crab-pot F1 trails the state of the art. DEPTH makes

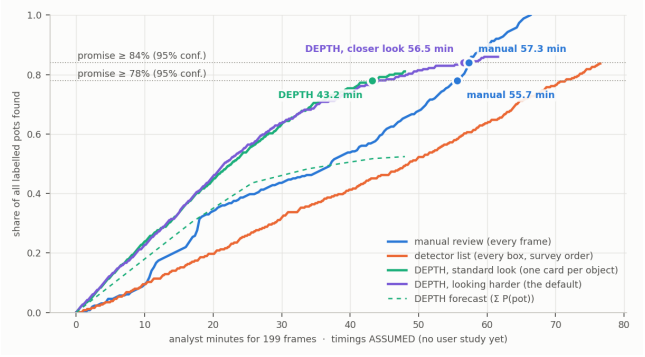


Figure 7: Share of pots found against analyst minutes on the 199 verification frames, assuming 20 s per frame for manual review and 8 s per card. Timings are assumptions until a user study is run.

Host (stock OpenCV 5)	Mode	ms/frame	FPS
c8g.xlarge Graviton4, 4 vCPU	standard	**138.5**	7.2
c8i.xlarge x86, 4 vCPU	standard	145.8	6.8
c8g.xlarge / c8i.xlarge	look harder	655 / 693	1.5 / 1.4
c8g.large Graviton4, 2 vCPU	std / harder	268 / 1,300	3.7 / 0.8
Graviton4, 1 core	standard	435	2.3

Table 7: Runtime (median per frame, full product path). The network forward pass accounts for 92–99% of the time.

the detector usable, not accurate. **Transfer.** A new sonar requires re-calibration; the applicability test detects a different sensor but is not designed for gradual drift in site or gain. **Guarantee scope.** The promise is a pooled rate fitted on three recordings and verified on three; one recording fell below it. **Field validation.** No recording has both labelled pots and real GPS, so positions on pot frames use labelled synthetic tracks. **Human factors.** Analyst times are assumed; the timed user study and blinded false-alarm audit are built but not yet run, and incomplete labels make measured precision a lower bound. **Failure modes.** On the earlier model, 32% of missed pots touched the frame edge (11% of found pots), and misses had lower contrast and longer range but were not smaller. **Sensing.** Pitch and roll are not corrected, data dropouts are flagged rather than repaired, and outputs are boxes rather than masks. **Deployment.** The COOL benchmark and edge hardware remain unmeasured.

8 Conclusion

DEPTH shows that an imperfect sonar detector can support operational decisions when it is wrapped in calibrated guarantees, an applicability test and measured geometry. The agent promises, with 95% confidence, that at least 78% of crab pots reach a person (84% when looking harder); both promises held on unseen recordings and survived a correction for correlated frames. It withdraws its promise where it would be false, places findings where the measurements put them, and acts only with human approval, all on a CPU with OpenCV 5. Because the calibration code is model-agnostic, a stronger detector can be substituted and its promise re-fitted with a

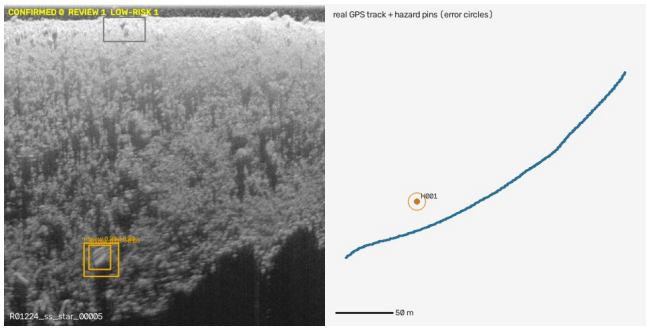


Figure 8: Raw Humminbird recording (Colorado River, 150.6 s, no known pots). Left: one review card. Right: real GPS track and the resulting pin with its 8.0 m error circle.

Idea	Outcome
Classical ROI gate	recall 0.708 → 0.147; retired
Shadow as a filter	precision 0.59 → 0.31–0.50; evidence only
Re-look re-scoring	no ranking gain; not used
Range-gain detector input	ΔAP −0.006 [−0.033, +0.019]
Seam inference	no recall gain
Larger input + flip TTA	+0.10 AP calib., −0.06 AP unseen
Active vision (VoI tool choice)	both gates failed; off by default
EXP-002/004/005 recipes	did not beat EXP-003

Table 8: Negative results, each from a registered study.

single command.

Acknowledgements. We thank the GhostVision and PING-Mapper authors for releasing their data. An AI coding assistant wrote much of the code under the author's direction; every number in this paper is produced by a script in the repository.

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